CHAPTER 1

Introduction to Machine Learning

Complete Exam-Oriented Notes

Syllabus covered: Human Learning, Machine Learning, Well-Posed Learning Problem, Supervised Learning, Classification, Regression, Unsupervised Learning, Clustering, Reinforcement Learning, Applications, Python, R and MATLAB.

Style: Simple English + required technical terminology + diagrams + examples + exam-ready definitions and comparisons.

General reference recommendation

For this chapter, a standard introductory Machine Learning textbook is useful for deeper reading, but the notes are designed to be self-contained for examination preparation. A commonly used reference is Machine Learning by Tom M. Mitchell, especially for the concept of the well-posed learning problem.

For practical Python-based ML, a useful reference is Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow by Aurélien Géron.

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1.1 Overview of Human Learning

Learning means gaining knowledge, skill or ability from experience, observation, study and practice. Humans learn continuously from their surroundings and use previous experience to deal with new situations.

How humans learn

• Observation: Learning by watching people, objects or events.

• Experience: Learning from what happens in real life.

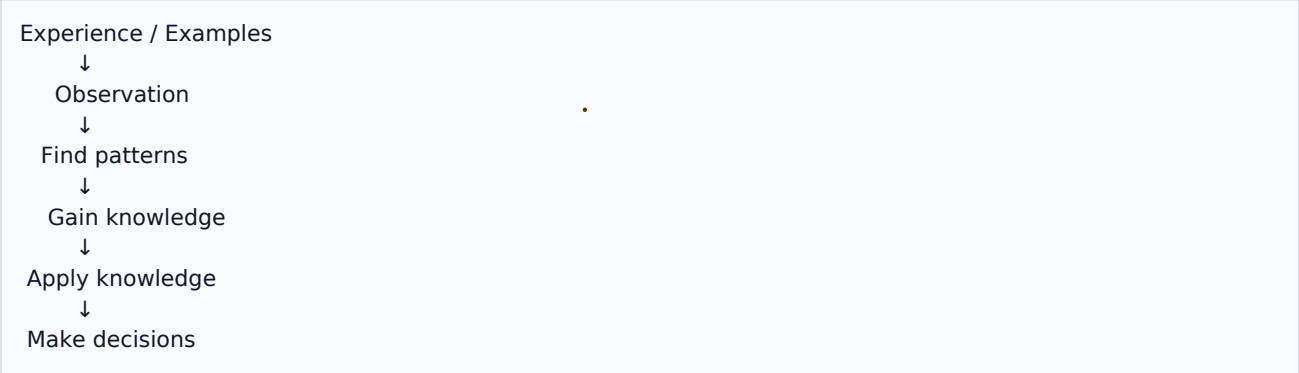
• Practice: Improving a skill through repeated activity.

• Examples: Understanding a new concept from known examples.

• Feedback: Learning from correct and incorrect results.

• Reasoning: Using existing knowledge to solve a new problem.

Simple learning process



Example

A child touches a hot object and feels pain. From this experience, the child learns that the object is hot and should not be touched. The child can later use this knowledge in a similar situation.

1.1.1 Difference between Human and Machine Learning

Human Learning	Machine Learning
Learns from experience, observation and practice.	Learns patterns from data and examples.
Can learn from a small number of examples in many situations.	Often requires a large amount of training data.
Can use common sense, reasoning and context.	Depends mainly on data, algorithms and the trained model.
Can transfer knowledge across many different tasks.	Usually designed for a particular task or group of tasks.
Can learn using multiple senses and real-world experience.	Learns from the data supplied to it.
Uses emotions and human understanding.	Does not have human emotions or human understanding.

Exam definition: Human learning is the process of acquiring knowledge or skills through experience, observation and practice, whereas machine learning is the process in which a computer system learns patterns from data and improves its performance for a task.

1.2 Overview of Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every possible situation.

Exam definition: Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data and improve their performance on a task without being explicitly programmed for every case.

Traditional Programming vs Machine Learning

```
Traditional Programming:
Rules / Program + Data → Computer → Output

Machine Learning:
Data + Expected Output → Learning Algorithm → Model → Prediction
```

Example: Spam Email Detection

In traditional programming, a programmer may write rules such as checking specific words. In machine learning, the system is trained using previously labelled spam and non-spam emails and learns patterns that help classify new emails.

1.2.1 How do Machines Learn?

A machine learns by using an algorithm to find useful patterns or relationships in data. The model is trained using available data and then tested on data it has not used for training.

```
        DATA
         ↓
Data Preparation
         ↓
Training Algorithm
         ↓
 Trained Model
         ↓
   Test Model
         ↓
Prediction / Decision
         ↓
   Feedback
         ↓
Improve Model
```

Main steps

1. Collect Data

Collect relevant information for the problem.

2. Prepare the Data

Clean data, handle missing or incorrect values, remove duplicates and select useful information.

3. Train the Model

Use a machine learning algorithm to learn patterns from training data.

4. Test the Model

Give unseen data to the model and compare predictions with actual results where applicable.

5. Make Predictions

Use the trained model on new input data.

6. Improve the Model

Use suitable changes to data, features, algorithm or parameters to improve performance.

1.2.2 Well-Posed Learning Problem

A well-posed learning problem clearly defines three things: Task (T), Performance Measure (P) and Experience (E).

Definition: A program is said to learn from experience E with respect to some task T and performance measure P if its performance at task T, as measured by P, improves with experience E.

```
T → Task
P → Performance Measure
E → Experience

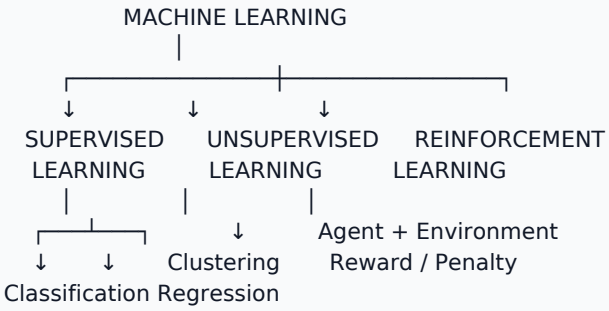
Memory trick: TPE = Task + Performance + Experience
```

Example: Email Spam Detection

Element	Example
Task (T)	Identify whether an email is spam or not spam.
Experience (E)	Previously classified emails used for training.
Performance (P)	Accuracy or another suitable measure of spam detection.

Exam point: The task, experience and performance measure must be clearly defined before calling a learning problem well-posed.

1.3 Types of Machine Learning



1.3.1 Supervised Learning

Supervised Learning is a machine learning technique in which a model learns from labelled training data. Both the input and the expected/correct output are available during training.

Definition: Supervised learning is a machine learning technique in which a model learns from labelled training data to predict the correct output for new input data.

```
Labelled Training Data
        ↓
   ML Algorithm
        ↓
   Trained Model
        ↓
   New Input Data
        ↓
     Prediction
```

Examples

- Spam / Not Spam email detection
- Disease / No Disease prediction

• House-price prediction

• Pass / Fail prediction

The two major supervised learning tasks in this syllabus are classification and regression.

1.3.1.1 Classification

Classification is a supervised learning technique used to assign an input to a predefined categorical class.

> Definition: Classification is a supervised learning technique in which the model assigns input data to one or more predefined classes or categories.

Input	Predicted Class
Email	Spam / Not Spam
Image	Cat / Dog
Transaction	Fraud / Genuine
Student result	Pass / Fail

Types

Binary classification: Two classes, such as Yes/No or Spam/Not Spam.

Multi-class classification: More than two classes, such as Cat/Dog/Horse/Cow.

Key point: Classification produces a discrete or categorical output.

1.3.1.2 Regression

Regression is a supervised learning technique used to predict a continuous numerical value.

> Definition: Regression is a supervised learning technique used to predict a continuous numerical output based on input data.

Examples include house-price prediction, temperature prediction, salary prediction and sales forecasting.

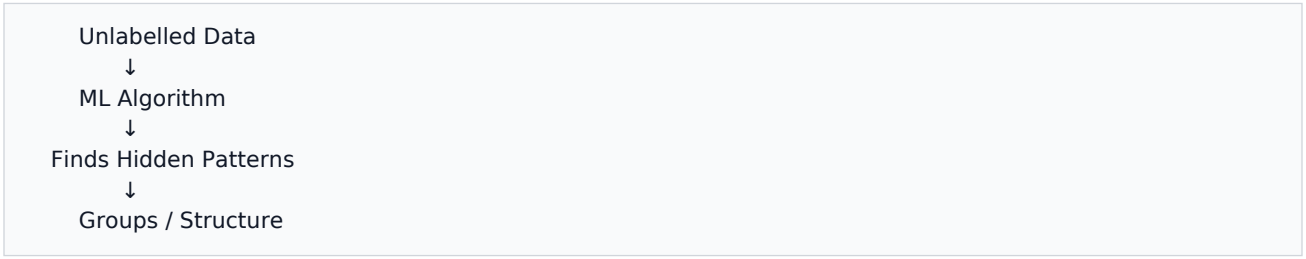
```
House Area
   ↓
Regression Model
   ↓
Predicted Price
   ↓
₹72.5 lakh
```

Classification	Regression
Predicts a category/class.	Predicts a numerical value.
Output is discrete/categorical.	Output is continuous numerical.
Example: Spam/Not Spam.	Example: House price.
Example: Pass/Fail.	Example: Temperature.

1.3.2 Unsupervised Learning

Unsupervised Learning is a machine learning technique in which the model learns from unlabelled data. There is no predefined correct output supplied during training.

> Definition: Unsupervised learning is a machine learning technique in which a model learns from unlabelled data to discover hidden patterns, structures or groups.

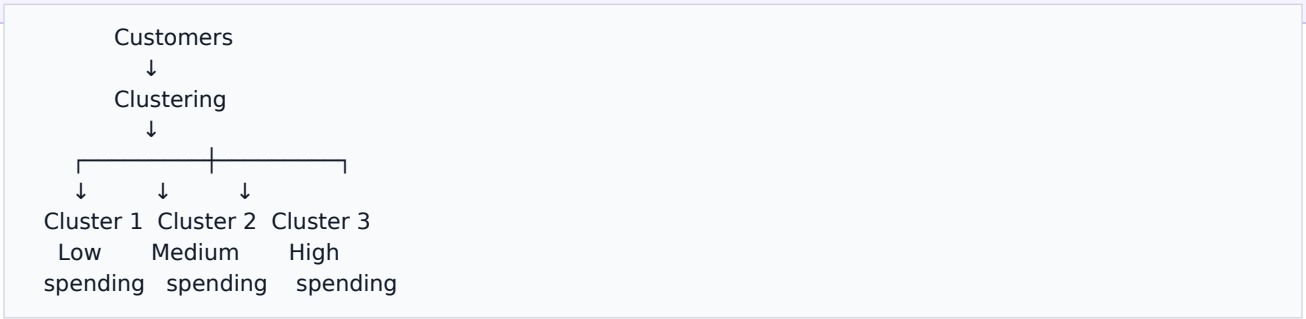


Common uses include customer segmentation, grouping similar documents, market analysis and finding unusual patterns.

1.3.2.1 Clustering

Clustering is an unsupervised learning technique that groups similar data objects together.

> Definition: Clustering is an unsupervised learning technique that divides data into groups called clusters, where objects within the same cluster are more similar to each other than to objects in other clusters.

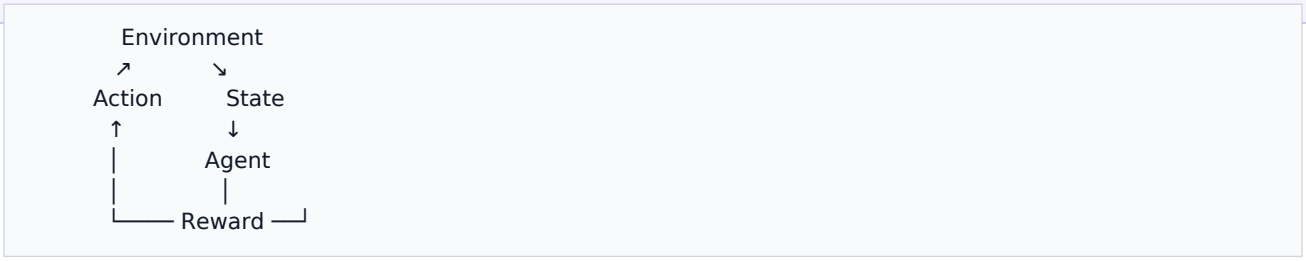


Important: In clustering, class labels are not provided beforehand. The algorithm discovers groups from similarities in the data.

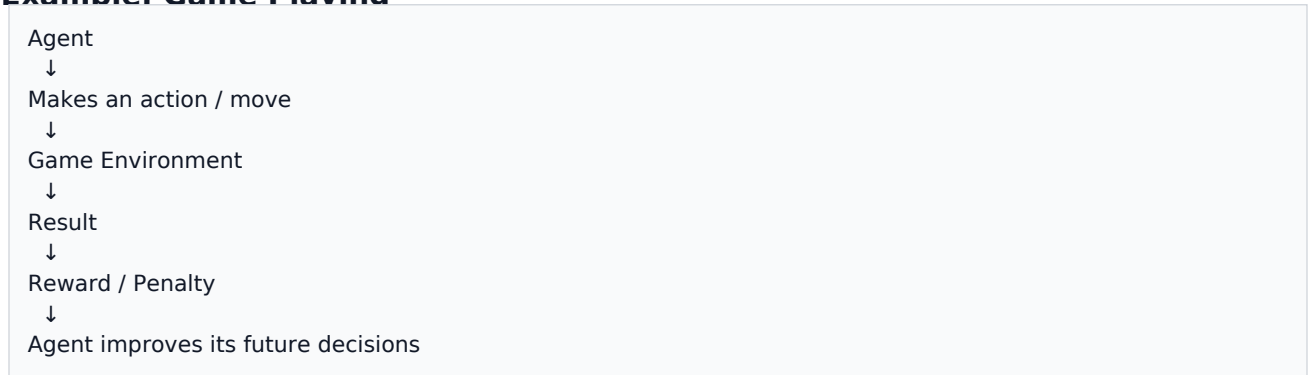
1.3.3 Reinforcement Learning

Reinforcement Learning (RL) is a type of machine learning in which an agent learns by interacting with an environment and receiving rewards or penalties for its actions.

> Definition: Reinforcement learning is a machine learning technique in which an agent learns to make decisions by interacting with an environment and receiving rewards or penalties for its actions.



Example: Game Playing



1.3.3.1 Elements of Reinforcement Learning

Element	Meaning
Agent	The learner or decision-making entity that takes actions.
Environment	The external system in which the agent operates and receives feedback.
State	The current condition or situation of the environment.
Action	An operation or decision performed by the agent.
Reward	A numerical feedback signal indicating how good or bad an action is.
Policy	A strategy that determines which action the agent should take for a given state.

Easy memory

Agent = learner | Environment = surroundings | State = current situation | Action = what agent does | Reward = feedback | Policy = strategy

1.3.4 Difference between Supervised, Unsupervised and Reinforcement Learning

Feature	Supervised	Unsupervised	Reinforcement
Training information	Labelled data	Unlabelled data	Interaction / experience
Correct output	Provided during training	Not provided	No direct correct output
Main objective	Predict output	Find hidden patterns/groups	Learn good actions
Feedback	Labels / target values	No direct target labels	Reward / penalty
Typical tasks	Classification, Regression	Clustering	Decision making
Example	Spam detection	Customer grouping	Game playing

Memory trick: Supervised = Teacher | Unsupervised = No Teacher | Reinforcement = Reward / Penalty

1.4 Applications of Machine Learning

1. Healthcare

Disease prediction, medical image analysis, patient risk prediction and drug discovery.

2. Finance and Banking

Fraud detection, credit scoring, risk analysis and financial forecasting.

3. E-commerce

Product recommendations, customer segmentation, demand prediction and personalized services.

4. Email

Spam detection and email classification.

5. Transportation

Traffic prediction, route optimization, driver assistance and autonomous vehicle systems.

6. Social Media

Content recommendation, image recognition, spam detection and personalized feeds.

7. Speech Recognition

Converting human speech into text and supporting voice-based systems.

8. Image Recognition

Face recognition, object detection, handwriting recognition and medical image analysis.

9. Agriculture

Crop disease detection, crop-yield prediction, weather analysis and soil analysis.

10. Manufacturing

Predictive maintenance, quality inspection, defect detection and production optimization.

1.5 Tools and Technology for Machine Learning

Common programming languages and environments used in machine learning include Python, R and MATLAB. Each has different strengths and is used according to the problem, development environment and requirements.

1.5.1 Python

Python is a high-level, general-purpose programming language widely used for machine learning, data science and artificial intelligence.

> Definition: Python is a high-level programming language widely used in machine learning because of its simple syntax, large library ecosystem and strong community support.

Library	Main purpose
NumPy	Numerical operations and arrays
Pandas	Data processing and analysis
Matplotlib	Data visualization
Scikit-learn	Machine learning algorithms
TensorFlow	Machine learning and deep learning
PyTorch	Machine learning and deep learning

Advantages

- Simple and readable syntax
- Large number of ML and data-science libraries
- Large developer community
- Open-source
- Suitable for rapid development

Simple workflow

```
Python
  ↓
NumPy / Pandas
  ↓
Data Processing
  ↓
Scikit-learn
  ↓
ML Model
```

↓
Prediction

1.5.2 R Programming Language

R is an open-source programming language and environment mainly designed for statistical computing, data analysis and data visualization.

Definition: R is an open-source programming language widely used for statistical analysis, data visualization and machine learning.

Package	Main purpose
ggplot2	Data visualization
dplyr	Data manipulation
caret	Machine learning
randomForest	Random forest algorithms
tidyr	Data cleaning and organization

Advantages

- Excellent statistical capabilities
- Strong data visualization
- Open-source
- Large package ecosystem
- Useful for research and data analysis

1.5.3 MATLAB

MATLAB stands for MATrix LABoratory. It is a numerical computing environment and programming language widely used for mathematical computation, data analysis, simulation, engineering and machine learning.

Definition: MATLAB is a numerical computing environment and programming language used for mathematical computation, data analysis, visualization, simulation and machine learning.

Uses

- Mathematical calculations
- Matrix operations
- Data analysis
- Simulation
- Signal processing
- Image processing
- Machine learning

Why MATLAB is useful in ML

MATLAB provides convenient mathematical operations, visualization and specialized toolboxes, making it useful for engineering, education, research and numerical applications.

Python vs R vs MATLAB

Feature	Python	R	MATLAB
Main strength	General programming + ML	Statistics + data analysis	Numerical computing + engineering
ML support	Excellent	Excellent	Good
Statistics	Very good	Excellent	Very good
Visualization	Excellent	Excellent	Good
Open source	Yes	Yes	Generally commercial
Typical use	ML, AI, software and data science	Statistics, research and data analysis	Engineering, mathematics and simulation

Exam-Ready Definitions

Machine Learning: Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and improve their performance on a task without being explicitly programmed for every case.

Supervised Learning: Supervised learning is a machine learning technique in which a model learns from labelled training data to predict the correct output for new input data.

Classification: Classification is a supervised learning technique in which input data is assigned to predefined classes or categories.

Regression: Regression is a supervised learning technique used to predict a continuous numerical value.

Unsupervised Learning: Unsupervised learning is a machine learning technique in which a model learns from unlabelled data to discover hidden patterns or structures.

Clustering: Clustering is an unsupervised learning technique that groups similar data objects into clusters.

Reinforcement Learning: Reinforcement learning is a machine learning technique in which an agent learns by interacting with an environment and receiving rewards or penalties for its actions.

Agent: An agent is the learner or decision-making entity in reinforcement learning that takes actions in an environment.

Environment: An environment is the external system with which the agent interacts in reinforcement learning.

State: A state represents the current situation or condition of the environment.

Action: An action is an operation or decision performed by the agent.

Reward: A reward is a numerical feedback signal received by the agent after performing an action.

Policy: A policy is a strategy that determines which action an agent should take for a given state.

Well-Posed Learning Problem: A well-posed learning problem clearly defines the Task (T), Performance measure (P) and Experience (E).

Quick Revision Sheet

```
MACHINE LEARNING
│
├── Supervised → Labelled data
```

```
│   ├── Classification → Category / class
│   └── Regression → Continuous numerical value
│
├── Unsupervised → Unlabelled data
│   └── Clustering → Groups similar objects
│
└── Reinforcement → Agent + Environment
    ├── State
    ├── Action
    ├── Reward
    └── Policy

Well-Posed Learning Problem:
T = Task
P = Performance measure
E = Experience

Memory:
Supervised = Teacher
Unsupervised = No Teacher
Reinforcement = Reward / Penalty

Tools:
Python → General ML / AI / programming
R → Statistics / data analysis
MATLAB → Mathematics / engineering / simulation
```

Important Exam Questions to Prepare

1. Define Machine Learning and explain how it differs from traditional programming.

2. Explain the well-posed learning problem with Task, Performance Measure and Experience using an example.

3. Explain the types of Machine Learning with a suitable diagram.

4. Explain Supervised Learning. Differentiate Classification and Regression.

5. Explain Classification with examples. Explain binary and multi-class classification.

6. Explain Regression with examples.

7. Explain Unsupervised Learning and Clustering.

8. Explain Reinforcement Learning with a suitable diagram.

9. Explain the elements of Reinforcement Learning: Agent, Environment, State, Action, Reward and Policy.

10. Differentiate Supervised, Unsupervised and Reinforcement Learning.

11. Explain applications of Machine Learning.

12. Explain Python, R and MATLAB as tools for Machine Learning.

13. Differentiate Python, R and MATLAB.

Last-Minute Memory Map

ML → 3 main types: Supervised, Unsupervised, Reinforcement.
Supervised → 2 tasks: Classification and Regression.
Unsupervised → Clustering: group similar data without labels.
Reinforcement → Agent interacts with Environment: State → Action → Reward → Learning.

Well-posed problem → TPE: Task + Performance + Experience.
Tools → Python, R, MATLAB.